

Reading and Comparing Multi-Digit Numbers

Answer Key

Grades 4–5 Mathematics

Quick Answers

- | | |
|----------|------------------------------|
| 1. 800 | 6. > |
| 2. 6000 | 7. 23045, 23450, 23504 |
| 3. 37409 | 8. 10 |
| 4. 62418 | 9. the correct symbol = <, — |
| 5. < | 10. (a) = >, (b) = 270 |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 9. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics

4.NBT.A.1 – *problems 1, 2, 8*

4.NBT.A.2 – *problems 3, 4, 5, 6, 7, 9, 10*

Difficulty 1–4 of 5 across 12 tagged checks.

- 1.** The place-value chart shows 5,872. What is the value of the digit 8?

Solution: The chart puts the 8 in the *hundreds* column. The value of a digit is the digit times its place: $8 \times 100 = 800$. 800

- 2.** What is the value of the digit 6 in 46,205?

Solution: Reading places from the right: 5 ones, 0 tens, 2 hundreds, 6 thousands, 4 ten-thousands. The 6 sits in the thousands place, so its value is $6 \times 1,000 = 6,000$. 6000

- 3.** Write the number in standard form: $30,000 + 7,000 + 400 + 9$.

Solution: Each addend fills one place: 3 ten-thousands, 7 thousands, 4 hundreds, 0 tens (no tens are listed, so a zero holds that place), 9 ones. $30,000 + 7,000 + 400 + 9 = 37,409$. 37409

- 4.** Write in standard form: sixty-two thousand, four hundred eighteen.

Solution: “Sixty-two thousand” is 62,000; “four hundred” is 400; “eighteen” is 18. Adding the parts: $62,000 + 400 + 18 = 62,418$. Note the zero holding the empty hundreds–tens boundary is built in when we add. 62418

5. Write $<$, $>$, or $=$: $46,205 \square 46,502$.

Solution: Both numbers have five digits, and the ten-thousands (4) and thousands (6) match. Compare hundreds: $2 < 5$, so the left number is smaller.

$$46,205 < 46,502$$

6. Write $<$, $>$, or $=$: $72,341 \square 72,314$.

Solution: Ten-thousands, thousands, and hundreds all match (7, 2, 3). Compare tens: $4 > 1$, so the left number is greater.

$$72,341 > 72,314$$

7. Order from least to greatest: 23,450; 23,045; 23,504.

Solution: All three share 23 thousands—ten-thousands... compare the hundreds place: $0 < 4 < 5$. So 23,045 is least, then 23,450, then 23,504.

$$23,045, 23,450, 23,504$$

8. In 44,175, the ten-thousands 4 is how many times the thousands 4?

Solution: The first 4 is worth 40,000 and the second is worth 4,000. Each place is worth 10 times the place to its right, and these are neighboring places: $40,000 \div 4,000 = 10$.

$$10$$

9. Maya says $9,876 > 12,345$. Correct symbol, and what did she forget?

Solution: Count digits first: 9,876 has four digits (thousands at most) while 12,345 has five (it reaches the ten-thousands), so 12,345 is automatically greater. Maya compared the leading digits 9 and 1 without first lining up the place values.

$$9,876 < 12,345$$

A full-credit explanation says that you compare the *number of digits* (or the greatest place) before comparing any digits, and that a five-digit number always beats a four-digit number.

10. Saturday: 68,742 fans; Sunday: 68,472 fans. (a) Greater day? (b) How many more?

Solution: (a) Same ten-thousands and thousands (6, 8); compare hundreds: $7 > 4$, so Saturday's 68,742 is greater:

$$68,742 > 68,472 \text{ (Saturday)}$$

(b) Subtract: $68,742 - 68,472 = 270$. (The thousands and ten-thousands cancel; $742 - 472 = 270$.)

$$270$$